

MediaTek Helio X27

Featuring an ultra-powerful 10 core processor, the Helio X27 was the chip that powered the last generation of smartphones and is still in use. The SoC was designed keeping phones with dual cameras in mind and comes with native support for colo and mono sensor, capable of capturing three times the light compared to other SoCs using traditional sensors.

The deca Core CPU features another embedded processor, ARM Cortex M4, to take care of always-on background applications like music player.

Thanks to the three cluster architecture, the SoC is power efficient, delegating tasks to different clusters depending on how intensive they are.

Primary CPU	dual-ARM-A72 @ 2.6GHz + quad-ARM-A53 @ 2.0GHz + quad-ARM-A53 @ 1.6GHz
CPU architecture	Tri-Cluster Architecture
CPU structure	RISC
DSP type	-
GPU	ARM Mali-T880 MP4
Year Released	2016
Devices	Elephone S7, S8, UMi Z, LeEco Le Pro 3

Qualcomm Snapdragon

Qualcomm is an American semiconductor and telecommunication equipment company that has its own range of specially designed SoCs for mobile products. Qualcomm was the first company to launch a SoC featuring a 1 GHz processor for mobile phones almost a decade back in the year 2007. The latest generation of Qualcomm processors are the state of the art 800 series which feature quad and octa core CPUs.

Snapdragon 835

The Snapdragon 835 is the next generation of SoC that powers Samsung's latest flagship model, the Galaxy S8. It the first 10nm processor made by Qualcomm, a trend that its competitors are following with their latest SoCs too. The new 10 nm process makes the chip upto 30% smaller and leads to a lower power consumption and a higher performance. Sporting an Octa core processor in the big.LITTLE architecture, 4 performance oriented cores are clocked at 2.45 GHZ while the remaining 4 cores are clocked at 1.8